

LESSON 7.9

WRITING EQUATIONS FOR REAL-WORLD CONTEXTS



LESSON INTRODUCTION

MA.8.AR.3.3 Given a table, graph, or written description of a linear relationship, write an equation in slope-intercept form.

LEARNING TARGETS

- I can write the equation of a linear relationship in slope-intercept form given a graph, table of values, or a written description for a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

WORKING TOWARD

MA.912.AR.2.2 Write a linear two-variable equation to represent relationships between quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

ESSENTIAL QUESTIONS

- What is an independent variable?
- What is a dependent variable?
- Why do variables need to be defined prior to writing equations?
- What is needed to write an equation in slope-intercept form?

LESSON NARRATIVE

The lesson focuses on writing equations for linear relationships in slope-intercept form from a real-world context. Students learn the importance of defining variables when writing equations to represent a context. Students write the slope-intercept form of equations using the slope and y-intercept found in written descriptions, graphs, or tables.

COMPONENT SUMMARY

7.9.1 WARM-UP (~5 MIN.)

Evaluate the accuracy of real-world equations

7.9.3 GUIDED PRACTICE (10-15 MIN.)

Practice writing equations in slope-intercept form from a table, graph, or written description of a real-world context

7.9.5 WRAP-UP (~5 MIN.)

Reflect on understanding of the unit learning targets

7.9.2 GUIDED INSTRUCTION (15 MIN.)

Learn to write equations representing real-world contexts, including appropriately defining variables

7.9.4 YOUR TURN! (10 MIN.)

Independently practice writing equations in slope-intercept form from a table, graph, or written description of a real-world context

RESOURCES

GLOSSARY

Key Terms:

- Defining variables

Additional Terms:

- Independent variable
- Dependent variable
- Slope
- y-intercept

MATERIALS

- (Optional) Calculators
- (Optional) Highlighters

ADDITIONAL PREPARATION

- Consider having highlighters for students to mark the text throughout the lesson, particularly in the 7.9.2 Guided Instruction and 7.9.3 Guided Practice.
- Have examples of independent and dependent variables ready to help students who struggle to see the difference.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly define variables as objects (i.e., let x represent the apples) instead of quantities (i.e. let x represent the number of apples).
- Students may struggle to find the slope or y-intercept.

THINGS TO CONSIDER

- Defining variables in 7.9.1 Warm-Up is crucial for students to be able to write equations in the subsequent components.
- Some students may benefit from a spiral review of slope and y-intercept.
- In 7.9.3 Guided Practice, question 2 does not include the scaffolded steps that are included in question 1. Consider including the questions to guide students who are stuck.
- Consider spiraling the interpretation of slope and y-intercept throughout the lesson for each context.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

7.9.1 Warm-Up positions students to compare two different equations that were written to represent a scenario. Consider applying the Compare and Connect routine to foster students' meta-awareness and support rich discussion while they reason about which equation is correct.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In 7.9.3 Guided Practice, students determine the slope and y-intercept from a graph, table, or written description and write equations that represent real-world situations. The focus on defining variables in this lesson provides students clear direction on how to model real-world contexts appropriately with equations.

DIFFERENTIATE INSTRUCTION

During the 7.9.2 Guided Instruction, use a Think-Aloud while marking the text. Sharing the reasoning behind circling and underlining terms and numbers will provide visual and auditory entry points.

In the 7.9.5 Wrap-Up, provide an alternative task for students who struggle to express their understanding of the topics in writing. Have students rate the questions from 7.9.3 Guided Practice and 7.9.4 Your Turn!. For example, 2 = fully understood, 1 = partially understood, 0 = did not understand.

7.9.1 WARM-UP: WHOM DO YOU AGREE WITH?

TEACHER MOVES

Position students to discover the importance of defining variables by finding students who could argue that either student is correct. Using their reasoning and logic, probe students to think about what would make one correct and the other incorrect. Challenge students to think about what could be added to their answers to make the choice clearer.

ENRICHMENT

Whom do you agree with and why? What would make one equation correct and the other incorrect?



7.9.1 Warm-Up: Whom Do You Agree With?

1. Danica pays for a live TV streaming service. The only costs associated with the service are a streaming fee of \$50.72 per month. She also had to buy a streaming device to access the service. The streaming device cost her \$29.99.

Two students wrote an equation to represent the linear relationship.

Cameron's Equation	Imani's Equation
$r = 50.72q + 29.99$	$q = 50.72r + 29.99$

- ☐ Only Imani
- ☐ Only Cameron
- ☐ Both Imani and Cameron
- ☒ Neither Imani nor Cameron

wrote the correct equation

because the variables have not been defined.

Answers will vary.



7.9.2 Guided Instruction: Writing Equations for Real-World Contexts

1. Consider the equations Cameron and Imani wrote to represent the situation in the Warm-Up.
 - a. What did both students do correctly when writing their equations?
 - *Correctly determined the rate of change from context and recognized it would be multiplied by a variable*
 - *Correctly determined the initial value and recognized it would be the constant term*

Neither student defined the variables for the situation, which makes it difficult to determine which equation accurately represents the context.



Defining Variables

When writing equations to represent real-world contexts, **any variables used must be defined** to identify which quantity each variable in the equation represents.

- b. If r is the number of months Danica continues the streaming service and q is the total cost to Danica, which student wrote the correct equation?
Imani $q = 50.72r + 29.99$
- c. In this context, which quantity is the independent variable?
 r , the number of months
- d. Which quantity is the dependent variable?
 q , the total cost

When writing equations in slope-intercept form, $y = mx + b$, x is the independent variable and y is the dependent variable. The value of y changes depending on the value of x .

2. The temperature at sea level near Mount Rainier in the state of Washington on a cloudless fall day was 14°C . At the base of Mount Rainier, 0.5 kilometers (km) above sea level, the temperature was 11°C . The temperature on the mountain 5 km above sea level was -16°C .
 - a. What are the two quantities being related in this context, including units?
temperature ($^{\circ}\text{C}$) and elevation (km)
 - b. Define each quantity with a variable.
 *e = the elevation in kilometers **
 t = temperature in $^{\circ}\text{C}$
 - c. Place an asterisk (*) next to the independent variable.
 - d. There is a linear relationship between the temperature and altitude. Find the slope and y-intercept for this relationship.

Slope	y-intercept
-6	14

(0,14)
(0.5,11)
(5,-16)

$$\frac{-16 - 11}{5 - 0.5} = \frac{-27}{4.5} = -6$$

- e. Write the equation of the linear relationship using the defined variables.

$$t = -6e + 14$$

7.9.2 GUIDED INSTRUCTION: WRITING EQUATIONS FOR REAL-WORLD CONTEXTS

TEACHER MOVES

While going over answers with students, explicitly model your reasoning by using a Think-Aloud.

ENRICHMENT

- What is an example of an independent variable?
- What is an example of a dependent variable?
- When given a scenario, how do you determine which is the independent and which is the dependent variable?

DIFFERENTIATE INSTRUCTION

For students who struggle working with real-world contexts, consider marking the text in question 5 while sharing aloud why parts of the text are being circled and underlined. It is important for students to hear the thinking process while also seeing it in action.

ENRICHMENT

- In question 5, what information is given to help determine the slope and y-intercept?
- How can you check your equation to ensure it is correct?

QUESTIONING

Challenge students once they complete question 5 to use their equation to find the temperature at the summit that day if the summit of Mount Rainier is 14.4 kilometers above sea level.

7.9.3 GUIDED PRACTICE: WRITING EQUATIONS FOR REAL-WORLD CONTEXTS

DIFFERENTIATE INSTRUCTION

Consider giving students the opportunity to work with a partner. Discussing questions with a partner will build student confidence and understanding.

ENRICHMENT

- What is the important information from the written description? How can you use that information to find the slope and y-intercept?
- What are the two quantities in the relationship?
- Which is the independent variable? Which is the dependent variable?

QUESTIONING

Challenge students to prove their equations are correct using a mathematical model or computation. For example, have students graph their lines using a graphing utility to show that it goes through the points described in the real-world context, or have students substitute values into their equation to ensure that it is mathematically accurate.



7.9.3 Guided Practice: Writing Equations for Real-World Contexts

- In 2015, the number of people with paid subscriptions to on-demand music streaming services was approximately 10.54 million. The number of people with paid subscriptions has been increasing at a constant rate since then. By 2019, the number of people with paid subscriptions to music streaming services increased to about 59.74 million.

- Define the quantities in this relationship using variables.

x = the number of years since 2015

y = the number of paid subscriptions (in millions)

- Find the slope and y-intercept for this relationship.

Slope	y-intercept
12.3	10.54

$$\frac{59.74 - 10.54}{4 - 0} = \frac{49.2}{4} = 12.3 \quad \text{*Ask for interpretation}$$

- Write an equation to represent the relationship between the number of people, in millions, with paid subscriptions to music streaming services, and the number of years since 2015.

$$y = 12.3x + 10.54$$

- Due to the increase in music streaming platforms, the revenue earned by the music industry from downloaded music content has decreased over the same period of time. The table shows the approximate revenues earned each year from 2015 to 2019, in millions of dollars.

Years since 2015	Revenue in Millions of Dollars
0	2,238
1	1,865
2	1,492
3	1,119
4	746

$$\frac{1865 - 2238}{1 - 0} = \frac{-373}{1} = -373$$

Work with your partner to write an equation that represents the relationship between the quantities in this context.

x = the number of years since 2015

r = the revenue earned from downloaded music (in millions)

$$r = -373x + 2238$$



7.9.4 Your Turn!

- When planes descend for landing, they descend at a slower rate once they reach 10,000 feet above the ground. Three minutes after starting the final descent, planes are approximately 5,530 feet above the ground.

Write a linear equation in slope-intercept form that represents the relationship between a plane's height above the ground and time on its final descent.

x = the number of minutes

y = the plane's height in feet

$$\frac{10,000 - 5,530}{0 - 3} = \frac{4470}{-3} = -1,490$$

$$y = -1,490x + 10,000$$

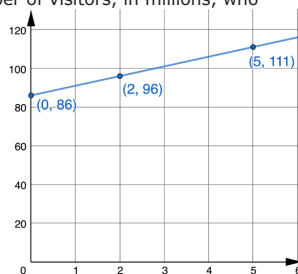
- Tourism is a major industry for Florida. The graph shows the total number of visitors, in millions, who traveled to Florida from 2011 to 2017.

Write a linear equation in slope-intercept form that represents this relationship.

x = the number of years since 2011

y = the number of tourists visiting Florida (in millions)

$$y = 5x + 86$$



$$\frac{96 - 86}{2 - 0} = \frac{10}{2} = 5$$

7.9.4 YOUR TURN!

ENRICHMENT

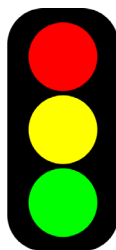
- Why did you choose _____ to represent the independent variable?
- Why did you choose _____ to represent the dependent variable?
- How did you check your equation to ensure it is correct?



7.9.5 Wrap-Up: Reflection

In this unit, you have learned multiple concepts about linear relationships, including those listed below.

- Finding the slope of a linear relationship
 - Finding the y -intercept of a linear relationship
 - Interpreting the slope and y -intercept in context
 - Graphing a linear relationship
 - Writing the equation of a linear relationship
- Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.



I still feel stuck on how to ...

I'm starting to get the hang of ...

I feel confident and ready to go with ...

Answers will vary.

7.9.5 WRAP-UP: REFLECTION

TEACHER MOVES

Consider providing an alternative task for students who struggle to express their understanding of the topics in writing. Have students rate the questions from 7.9.3 Guided Practice and 7.9.4 Your Turn!. For example, 2 = fully understood, 1 = partially understood, and 0 = did not understand. Use this formative assessment data to provide remediation to struggling students.